Arista Access Node Optical Transport Failure Resolution Procedure for KR-ACC-24

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_7798

Description

The KR-ACC-24 node, situated at the KR Puram Access site and operating as an access role, has encountered an Optical Transport Failure, denoted by the ALARM_7798 warning. This failure domain is precipitated by degradation or failure of SFP/XFP transceivers, fiber cuts, or significant loss of receive/transmit power. The underlying mechanics involve the physical layer of the optical transport, where the integrity of the signal transmission is compromised. This can be due to various factors including but not limited to, aging of the optical components, physical damage to the fibers, or misconfiguration of the optical interfaces. The Arista equipment, with its CLI dialect, provides a robust set of commands to diagnose and potentially resolve these issues. Understanding the protocol and hardware mechanics is crucial; for instance, the use of Ethernet protocols over optical interfaces requires precise configuration and monitoring to ensure seamless data transmission. The failure in this domain can lead to significant network downtime and data loss, emphasizing the need for a swift and accurate resolution process. This SOP aims to guide through the diagnostic and resolution process, leveraging Arista's CLI commands to address the Optical Transport Failure on KR-ACC-24 efficiently.

Pre-requisite Checks

- Dispatch remote hands to KR Puram Access for physical inspection and intervention if necessary.

- Verify the status of redundant links to ensure network resilience.

- Confirm the power supply and environmental conditions are within acceptable parameters for the Arista equipment.

Potential Root Causes

Cause	Probability
SFP/XFP Transceiver Degradation	High
Fiber Cuts or Physical Damage	Medium
Rx/Tx Power Loss Due to Misconfiguration	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces transceiver, Expected Output: Detailed transceiver information including vendor, model, and current operating parameters., Description: This command isolates issues in the physical layer by providing insights into the transceiver's health and performance, useful for identifying degradation or failure.
```

```
Command: show interfaces ethernet counters, Expected Output:
Interface-specific counters for errors, discards, and other
performance metrics., Description: This command helps in diagnosing
data plane issues by analyzing interface counters, which can indicate
signal quality problems or configuration errors.
```

Resolution Steps

Step 1: Step: Replace Faulty Transceiver, Command: replace transceiver /, Description: Physically replacing a degraded or faulty transceiver with a new one, ensuring compatibility with the Arista equipment and the specific interface.

Step 2: Step: Adjust Optical Power Levels, Command: interface ethernet optical-power , Description: Configuring the optical power levels to match the requirements of the connected device or the optical link, ensuring optimal signal transmission.

Post-Check Verification

```
Command: show interfaces transceiver detail, Expected Output: Detailed
information about the transceiver, including its operating parameters
and any alarms., Description: Verifying that the transceiver
replacement or configuration adjustments have resolved the issues,
with the transceiver operating within specified parameters.

Command: show interfaces ethernet counters detail, Expected Output:
Detailed interface counters, including errors, discards, and other
performance metrics., Description: Confirming that the data plane
issues have been resolved, with interface counters indicating healthy
traffic flow and minimal to no errors.
```

Roll-back Procedures

- Step: Revert to Original Transceiver, Command: replace transceiver / with original, Description: If the replacement transceiver does not resolve the issue, reverting to the original transceiver to ensure the problem is not with the new component.

- Step: Reset Optical Power Levels, Command: default interface ethernet optical-power, Description: Resetting the optical power levels to their default values if adjustments did not resolve the issue, to prevent any further configuration-related problems.